MAST20030 Differential Equations
Problem Sheet 1: Linear ODEs and IVPs

1. Find the general solutions to the following ODEs:

   (a) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + t^2 y = 0$

   (b) $\dfrac{\mathrm{d}y}{\mathrm{d}t} - 2ty = 1$

   (c) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{y}{t} = 2 + t^2$

   (d) $(t^2+1)\dfrac{\mathrm{d}y}{\mathrm{d}t} - 2ty = t^2 + 1$

   (e) $t\dfrac{\mathrm{d}y}{\mathrm{d}t} + (t+1)y = 3\mathrm{e}^{-t}$

   (f) $t\dfrac{\mathrm{d}y}{\mathrm{d}t} + 2y = \mathrm{e}^t + \log(t)$

   (g) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + \tan(t)y = \sec(t)$

2. Find the unique solution of the following initial value problems and state the intervals in which the solutions are valid:

   (a) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + 2y = t + \sin(t), \qquad y(0) = 4$

   (b) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + ty = t^3, \qquad y(0) = 3$

   (c) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{4y}{t+2} = 3, \qquad y(0) = -5$

   (d) $(2t^2+1)\dfrac{\mathrm{d}y}{\mathrm{d}t} - 10ty = 4t, \qquad y(0) = 7$

   (e) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + y = \mathrm{e}^{-2t}, \qquad y(0) = 1$

   (f) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{2y}{t} = \dfrac{1}{t^2}, \qquad y(t_0) = y_0$

   (g) $\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{y}{t^{1/3}} = 0, \qquad y(t_0) = y_0$

3. By trialling solutions of the form $y = \mathrm{e}^{\lambda t}$, find the general solutions of the following homogeneous ODEs:

   (a) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}t} + 3y = 0$

   (b) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\dfrac{\mathrm{d}y}{\mathrm{d}t} + 4y = 0$

   (c) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\dfrac{\mathrm{d}y}{\mathrm{d}t} - 3y = 0$

   (d) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}t} + 13y = 0$

   For each of these ODEs:

   i. Trial exponential solutions of the form $y = \mathrm{e}^{\lambda t}$ to find the general solution.

   ii. Rewrite the system as a first order vector-valued ODE, i.e., in matrix form.

   iii. Find the eigenvalues and eigenvalues of the matrix, and thereby find the general solution to the vector-valued ODE.

   iv. Draw the phase portrait.

   Now find the general solutions to the following inhomogeneous ODEs:

   (e) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}t} + 3y = 3\mathrm{e}^t$

   (f) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - 5\dfrac{\mathrm{d}y}{\mathrm{d}t} + 4y = 10\sinh(t)$

   (g) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2\dfrac{\mathrm{d}y}{\mathrm{d}t} - 3y = 2 + 3\mathrm{e}^t$

   (h) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4\dfrac{\mathrm{d}y}{\mathrm{d}t} + 13y = \sin(t)$

4. By trialling solutions of the form $y = t^n$, find all solutions of the following ODEs:

   (a) $t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 7t\dfrac{\mathrm{d}y}{\mathrm{d}t} + 9y = \log(t)$

   (b) $t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\dfrac{\mathrm{d}y}{\mathrm{d}t} - y = t$

   (c) $t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - 4t\dfrac{\mathrm{d}y}{\mathrm{d}t} + 6y = t^4\mathrm{e}^t$

   (d) $2t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 5t\dfrac{\mathrm{d}y}{\mathrm{d}t} + y = t(t-1)$

   (e) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - \dfrac{t+2}{t}\dfrac{\mathrm{d}y}{\mathrm{d}t} + \dfrac{t+2}{t^2}y = 0$

5. For the following ODEs, show that the function y_c is a complementary solution and use it to find the general solution:

   (a) $y_c = t - 1, \qquad (2t - t^2)\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 2(t-1)\dfrac{\mathrm{d}y}{\mathrm{d}t} - 2y = 1$

   (b) $y_c = t, \qquad t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\dfrac{\mathrm{d}y}{\mathrm{d}t} - y = t$

   In both cases, find the unique solution satisfying $y(1) = 0$ and $y'(1) = 0$. What happens if we replace the initial conditions by $y(0) = a$ and $y'(0) = b$, for some $a, b \in \mathbb{R}$?

6. We study the inhomogeneous second-order linear ODE

$$t^2\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + t\frac{\mathrm{d}y}{\mathrm{d}t} + y = 1.$$

   (a) Show that $v(t) = \cos(\log t)$ is a complementary solution for $t > 0$.

   (b) Use reduction of order to deduce the general solution to the inhomogeneous ODE above.

   (c) Explain how trialling $y = t^n$ could have led directly to the complementary solutions.

7. Find the general solutions of the following higher-order linear ODEs:

   (a) $\dfrac{\mathrm{d}^3y}{\mathrm{d}t^3} + 6\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 12\dfrac{\mathrm{d}y}{\mathrm{d}t} + 8y = 0$

   (b) $t\dfrac{\mathrm{d}^3y}{\mathrm{d}t^3} - \dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} = 0$

   (c) $\dfrac{\mathrm{d}^4y}{\mathrm{d}t^4} + 2\dfrac{\mathrm{d}^3y}{\mathrm{d}t^3} + \dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} = 0$

   (d) $\dfrac{\mathrm{d}^4y}{\mathrm{d}t^4} - 5\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 4y = 0$

   (e) $\dfrac{\mathrm{d}^3y}{\mathrm{d}t^3} + \dfrac{\mathrm{d}y}{\mathrm{d}t} = 6t$

   (f) $\dfrac{\mathrm{d}^3y}{\mathrm{d}t^3} + 3\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 3\dfrac{\mathrm{d}y}{\mathrm{d}t} + y = 4\mathrm{e}^{-t}$

   Don't forget to check that your complementary solutions are linearly independent!

8. (a) Find the eigenvalues and eigenvectors of the matrix

$$\begin{bmatrix} 4 & -3 \\ 5 & -4 \end{bmatrix}$$

   Use MATLAB, Mathematica, or any other computational package to verify your answers.

(b) Using part (a), find the general solution of the system

$$\frac{\mathrm{d}x}{\mathrm{d}t} = 4x - 3y, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = 5x - 4y$$

(c) Verify your answer to part (b) by showing that it satisfies the original system of differential equations.

9. (a) Using eigenvalues and eigenvectors, find the general solution to

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x - y, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = x + y$$

(b) Verify your answer to part (a) by showing that it satisfies the original system of differential equations.

10. Using eigenvalues and eigenvectors, attempt to find the general solution of the following coupled differential equations. What goes wrong?

(a) $\begin{aligned} \dot{x} &= x + 9y \\ \dot{y} &= -x - 5y \end{aligned}$ (b) $\begin{aligned} \dot{x} &= x + y \\ \dot{y} &= -x + 3y \end{aligned}$

11. Using eigenvalues and eigenvectors, solve the following systems of first order differential equations subject to the given initial conditions.

(a) $\begin{aligned} \dot{x} &= x + y \\ \dot{y} &= 4x - 2y \\ x(0) &= 1 \\ y(0) &= 2 \end{aligned}$ (b) $\begin{aligned} \dot{x} &= x + 2y \\ \dot{y} &= -\tfrac{1}{2}x + y \\ x(0) &= 2 \\ y(0) &= 3 \end{aligned}$

12. Rewrite the following systems of ODEs as vector-valued first-order linear ODEs and find their general solutions using matrix diagonalisation:

(a)
$$\begin{Bmatrix} f'(t) - g'(t) - f(t) = \cos t, \\ g'(t) - 2f(t) = \mathrm{e}^t \end{Bmatrix}$$

(b)
$$\begin{Bmatrix} f'(t) - 2g'(t) - g(t) = t^2, \\ f'(t) - 3g(t) = 0 \end{Bmatrix}$$

(c)
$$\begin{Bmatrix} f'(t) = -f(t) + g(t), \\ g'(t) = 2f(t) + 2g(t) + 2h(t), \\ h'(t) = g(t) - h(t) \end{Bmatrix}$$

13. Find the unique solution of the following IVPs using matrix diagonalisation:

(a)
$$\begin{Bmatrix} f'(t) = -2f(t) + g(t), & f(0) = 3, \\ g'(t) = -3f(t) + 2g(t) + 2\sin t, & g(0) = 4 \end{Bmatrix}$$

(b)

$$\left\{\begin{array}{ll} f'(t) = -f(t) - 3g(t) + 2h(t), & f(0) = -3, \\ g'(t) = -2g(t), & g(0) = 0, \\ h'(t) = -g(t) - 2h(t), & h(0) = 3 \end{array}\right\}$$

14. Find the solution to the following IVPs using matrix exponentials:

(a)

$$\left\{\begin{array}{ll} f'(t) = -2f(t) + g(t), & f(0) = a, \\ g'(t) = -4f(t) + 3g(t), & g(0) = b \end{array}\right\}$$

(b)

$$\left\{\begin{array}{ll} f'(t) = -2g(t), & f(0) = a, \\ g'(t) = 2f(t), & g(0) = b \end{array}\right\}$$

(c)

$$\left\{\begin{array}{ll} f'(t) = 3f(t) + g(t) - 2h(t), & f(0) = a, \\ g'(t) = -f(t) + 2g(t) + h(t), & g(0) = b, \\ h'(t) = 4f(t) + g(t) - 3h(t), & h(0) = c \end{array}\right\}$$

15. (a) Show that the wronskian of n *linearly dependent* (sufficiently differentiable) functions vanishes.
Hint: write down the linear dependence relation, differentiate it sufficiently many times and rewrite the result in terms of the matrix whose determinant is the wronskian. What does the result say about this matrix?

(b) The contrapositive of the previous statement is: if the wronskian of n (sufficiently differentiable) functions is non-zero, then the functions are linearly independent. This is the test we have used to check if complementary solutions are linearly independent. Show that the converse to this statement is false by verifying that

$$y_1(t) = t^2 \qquad \text{and} \qquad y_2(t) = t|t|$$

are differentiable for all $t \in \mathbb{R}$, are linearly independent as functions on $\mathbb{R}$, and have vanishing wronskian.
This counterexample is due to Peano who also proved that the converse is true if the functions are analytic.

16. The definition for the wronskian of the solutions of a vector-valued first-order linear ODE given in class does not look the same as the definition given for general higher-order linear ODEs. Show that upon converting such a higher-order ODE into a vector-valued first-order linear ODE, the definition for the former becomes the definition given for the latter. Hint: try showing this for a 2-dimensional system first before tackling the general n-dimensional case.

MAST20030 Differential Equations
Solutions to Problem Sheet 1: Linear ODEs and IVPs

1. (a) $y = Ce^{-\frac{t^3}{3}}$

   (b) $y = Ce^{t^2} + e^{t^2}\displaystyle\int_0^t e^{-\tau^2}\,d\tau$

   (c) $y = \dfrac{C}{t} + \dfrac{t^3}{4} + t$

   (d) $y = (1+t^2)(C + \arctan t)$

   (e) $y = \dfrac{1}{t}(C+3t)e^{-t}$

   (f) $y = \dfrac{C}{t^2} - \dfrac{1}{4} + \dfrac{(t-1)e^t}{t^2} + \dfrac{\log t}{2}$

   (g) $y = C\cos t + \sin t$

2. (a) $y = \dfrac{t}{2} - \dfrac{1}{4} + \dfrac{89}{20}e^{-2t} - \dfrac{1}{5}\cos t + \dfrac{2}{5}\sin t$, $t \in \mathbb{R}$

   (b) $y = t^2 - 2 + 5e^{-\frac{t^2}{2}}$, $t \in \mathbb{R}$

   (c) $y = \dfrac{1}{5}\left(6 + 3t - \dfrac{496}{(t+2)^4}\right)$, $t > -2$

   (d) $y = \dfrac{1}{5}\left(-2 + 37(2t^2+1)^{5/2}\right)$, $t \in \mathbb{R}$

   (e) $y = 2e^{-t} - e^{-2t}$, $t \in \mathbb{R}$

   (f) $y = \dfrac{1}{t} + \dfrac{t_0(t_0y_0 - 1)}{t^2}$, $t > 0$ if $t_0 > 0$, $t < 0$ if $t_0 < 0$ and no solution if $t_0 = 0$

   (g) $y = y_0 e^{-\frac{3}{2}(t^{2/3} - t_0^{2/3})}$, $t \in \mathbb{R}$

3. (a) $y = C_1e^{-3t} + C_2e^{-t}$

   (b) $y = C_1e^{t} + C_2e^{4t}$

   (c) $y = C_1e^{-3t} + C_2e^{t}$

   (d) $y = e^{-2t}(C_1\cos 3t + C_2\sin 3t)$

   (e) $y = C_1e^{-3t} + C_2e^{-t} + \dfrac{3}{8}e^t$

   (f) $y = C_1e^{t} + C_2e^{4t} - \dfrac{1}{2}e^{-t} - \dfrac{5}{9}e^t(1+3t)$

   (g) $y = C_1e^{-3t} + C_2e^{t} - \dfrac{2}{3} + \dfrac{3}{16}e^t(4t-1)$

   (h) $y = e^{-2t}(C_1\cos 3t + C_2\sin 3t) + \dfrac{1}{40}(3\sin t - \cos t)$

4. (a) $y = \dfrac{C_1}{t^3} + \dfrac{3C_2}{t^3}\log t + \dfrac{1}{27}(3\log t - 2)$

   (b) $y = \dfrac{C_1}{t} + C_2t + \dfrac{t}{4}(2\log t - 1)$

   (c) $y = C_1t^2 + C_2t^3 + t^2e^t$

   (d) $y = \dfrac{C_1}{t} + \dfrac{C_2}{t^{1/2}} + \dfrac{t}{30}(2t-5)$

   (e) $y = C_1t + C_2te^t$

5. (a) $y = C_1(t-1) + C_2(t-2)^2 - \dfrac{1}{2}$ and $y = \dfrac{1}{2}(t^2 - 2t + 1)$

   (b) $y = \dfrac{C_1}{t} + C_2t + \dfrac{1}{4}(2t\log(t) - t)$ and $y = \dfrac{1}{4t}(2t^2\log(t) - t^2 + 1)$

   In both cases, if we replace the initial conditions by $y(0) = a$ and $y'(0) = b$, for some $a, b \in \mathbb{R}$ then there is no solution.

6. (a) Substitute $v(t) = \cos(\log t)$ into the ODE.

(b) $y = C_1 \cos(\log t) + C_2 \sin(\log t) + 1$

(c) $n = \pm i$

7. (a) $y = C_1 \mathrm{e}^{-2t} + C_2 t \mathrm{e}^{-2t} + C_3 t^2 \mathrm{e}^{-2t}$

(b) $y = C_1 t^3 + C_2 t + C_3$

(c) $y = C_1 \mathrm{e}^{-t} + C_2 t \mathrm{e}^{-t} + C_3 t + C_4$

(d) $y = C_1 \mathrm{e}^{-2t} + C_2 \mathrm{e}^{-t} + C_3 \mathrm{e}^{t} + C_4 \mathrm{e}^{2t}$

(e) $y = C_1 \cos t + C_2 \sin t + C_3 + 3t^2$

(f) $y = C_1 \mathrm{e}^{-t} + C_2 t \mathrm{e}^{-t} + C_3 t^2 \mathrm{e}^{-t} + \dfrac{2}{3} t^3 \mathrm{e}^{-t}$

8. (a) $\lambda_1 = -1 \quad \boldsymbol{v}_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \qquad \lambda_2 = 1 \quad \boldsymbol{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = C_1 \begin{bmatrix} 3 \\ 5 \end{bmatrix} \mathrm{e}^{-t} + C_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \mathrm{e}^{t}$

9. (a) $x(t) = \mathrm{e}^t \left(C_1 \cos t - C_2 \sin t \right), \qquad y(t) = \mathrm{e}^t \left(C_1 \sin t + C_2 \cos t \right)$

10. In both cases, we can only find one linearly independent eigenvector so our trial solution does not lead to the general solution. The resolution is to use generalised eigenvectors or matrix exponentials which are not examinable. The single solution in each case is

(a) $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = C_1 \mathrm{e}^{-2t} \begin{bmatrix} -3 \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = C_1 \mathrm{e}^{2t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

11. (a) $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \dfrac{1}{5} \left(\mathrm{e}^{-3t} \begin{bmatrix} -1 \\ 4 \end{bmatrix} + 6 \mathrm{e}^{2t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$

(b) $\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \mathrm{e}^t \left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} \cos t + \begin{bmatrix} 6 \\ -1 \end{bmatrix} \sin t \right)$

12. (a) $\begin{bmatrix} f(t) \\ g(t) \end{bmatrix} = \dfrac{1}{3} C_1 \begin{bmatrix} 3\mathrm{e}^{3t} \\ 2\mathrm{e}^{3t} - 2 \end{bmatrix} + C_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \dfrac{1}{10} \begin{bmatrix} -5\mathrm{e}^t - 3\cos t + \sin t \\ -2\cos t - 6\sin t \end{bmatrix}$

(b) $\begin{bmatrix} f(t) \\ g(t) \end{bmatrix} = C_1 \begin{bmatrix} 3\mathrm{e}^t - 3 \\ \mathrm{e}^t \end{bmatrix} + C_2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \dfrac{1}{2} \begin{bmatrix} t^3 + 3t^2 + 6t + 6 \\ t^2 + 2t + 2 \end{bmatrix}$

(c) $\begin{bmatrix} f(t) \\ g(t) \\ h(t) \end{bmatrix} = C_1 \mathrm{e}^{-t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + C_2 \mathrm{e}^{-2t} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + C_3 \mathrm{e}^{3t} \begin{bmatrix} 1 \\ 4 \\ 1 \end{bmatrix}$

13. (a) $\begin{bmatrix} f(t) \\ g(t) \end{bmatrix} = 2\mathrm{e}^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \mathrm{e}^t \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \cos t - \begin{bmatrix} 1 \\ 2 \end{bmatrix} \sin t$

(b) $f(t) = -6\mathrm{e}^{-2t} + 3\mathrm{e}^{-t}, g(t) = 0, h(t) = 3\mathrm{e}^{-2t}$

14. (a) $\begin{bmatrix} f(t) \\ g(t) \end{bmatrix} = \dfrac{1}{3} \left(\begin{bmatrix} 4a - b \\ 4a - b \end{bmatrix} \mathrm{e}^{-t} + \begin{bmatrix} -a + b \\ -4a + 4b \end{bmatrix} \mathrm{e}^{2t} \right)$

(b) $f(t) = a \cos 2t - b \sin 2t, \qquad g(t) = b \cos 2t + a \sin 2t$

(c) $$\begin{bmatrix} f(t) \\ g(t) \\ h(t) \end{bmatrix} = \frac{1}{6}\left(\begin{bmatrix} -7a+7c \\ 2a-2c \\ -13a+13c \end{bmatrix} \mathrm{e}^{-t} + \begin{bmatrix} 15a-6b-9c \\ 0 \\ 15a-6b-9c \end{bmatrix} \mathrm{e}^{t} + \begin{bmatrix} -2a+6b+2c \\ -2a+6b+2c \\ -2a+6b+2c \end{bmatrix} \mathrm{e}^{2t} \right)$$

15. 16. Solutions to proofs not provided.